

PE-EC802A — Mixed Signal Design

Semester VIII • 3L:0T:0P • 3 Credits

Module 5 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Phase-Locked Loop (PLL) Architecture & Working

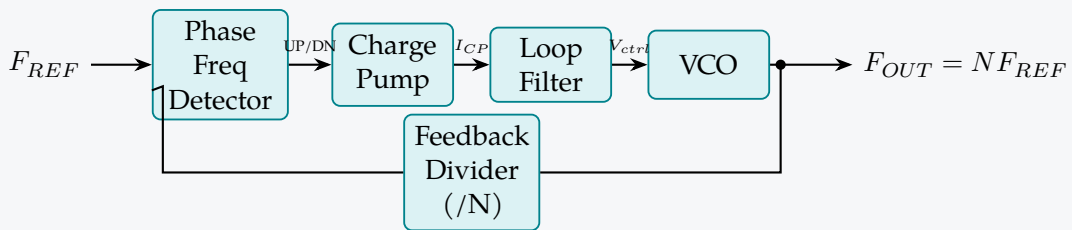
Phase-Locked Loop (PLL) Concept

A **Phase-Locked Loop (PLL)** is a feedback control system that generates an output signal whose phase is locked to the phase of an input reference signal. In mixed-signal systems, PLLs are widely used for frequency synthesis, clock recovery, and jitter reduction.

1.1 Block Components and Signal Flow

A Charge-Pump Phase-Locked Loop (CP-PLL) consists of five main blocks:

Block Diagram of a Charge-Pump PLL



Working of Each Block:

- 1. Phase Frequency Detector (PFD):** Compares the reference clock F_{REF} with the feedback clock F_{FB} . It generates digital pulses (UP and DN) whose widths are proportional to the phase and frequency difference between the inputs.
- 2. Charge Pump (CP):** Converts the digital UP/DN control pulses into a proportional current charge injection:

$$I_{avg} = I_{CP} \frac{\Delta\Phi}{2\pi}$$

- 3. Loop Filter (LF):** A passive low-pass filter (usually a resistor R in series with a capacitor C_1 , in parallel with a smaller capacitor C_2). It integrates the CP current to generate a smooth analog tuning voltage V_{ctrl} .
- 4. Voltage Controlled Oscillator (VCO):** A circuit whose output frequency is controlled by V_{ctrl} . The output frequency is:

$$\omega_{out} = \omega_{fr} + K_{VCO} V_{ctrl}$$

where ω_{fr} is the free-running frequency and K_{VCO} is the VCO gain (rad/s/V).

- 5. Feedback Divider (/N):** Divides the high output frequency down to the reference frequency, allowing the loop to lock at $F_{OUT} = N F_{REF}$.

1.2 Mathematical Loop Dynamics (Type-II CP-PLL)

A Type-II PLL has two integrators in the loop: the VCO (which integrates frequency to phase: $\Phi_{out} = \int \omega_{out} dt \Rightarrow H_{VCO}(s) = K_{VCO}/s$) and the loop filter (which contains an integrating capacitor).

Transfer Functions of Components:

- PFD/CP Gain: $K_d = \frac{I_{CP}}{2\pi}$ (A/rad).
- Loop Filter Impedance: $Z_{LF}(s) = R + \frac{1}{sC_1} = \frac{1+sRC_1}{sC_1}$ (ignoring the small parallel capacitor C_2).
- VCO Transfer Function: $H_{VCO}(s) = \frac{K_{VCO}}{s}$.

Open-Loop Gain $A(s)$:

$$A(s) = K_d \cdot Z_{LF}(s) \cdot \frac{H_{VCO}(s)}{N} = \left(\frac{I_{CP}}{2\pi} \right) \left(\frac{1+sRC_1}{sC_1} \right) \left(\frac{K_{VCO}}{s} \right) \frac{1}{N}$$

$$A(s) = \frac{I_{CP}K_{VCO}(1+sRC_1)}{2\pi NC_1 s^2}$$

Closed-Loop Transfer Function $H_{cl}(s) = \frac{\Phi_{out}(s)}{\Phi_{in}(s)}$:

$$H_{cl}(s) = \frac{N \cdot A(s)}{1 + A(s)} = \frac{\frac{I_{CP}K_{VCO}R}{2\pi} s + \frac{I_{CP}K_{VCO}}{2\pi C_1}}{s^2 + \frac{I_{CP}K_{VCO}R}{2\pi N} s + \frac{I_{CP}K_{VCO}}{2\pi NC_1}}$$

Comparing this with the standard second-order system denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$:

Type-II CP-PLL Dynamics

Natural Frequency (ω_n):

$$\omega_n = \sqrt{\frac{I_{CP}K_{VCO}}{2\pi NC_1}}$$

Damping Factor (ζ):

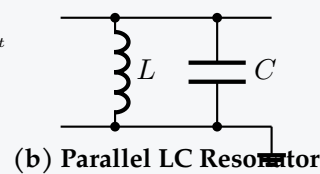
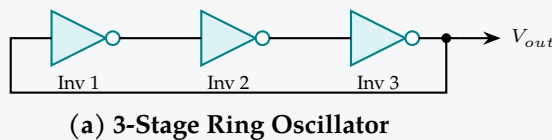
$$\zeta = \frac{RC_1}{2} \omega_n = \frac{R}{2} \sqrt{\frac{I_{CP}K_{VCO}C_1}{2\pi N}}$$

The resistor R is critical; it introduces a zero in the open-loop transfer function, stabilizing the second-order feedback loop by providing adequate phase margin (damping).

1.3 VCO Design: Ring Oscillator vs. LC-Tank

The VCO is the most noise-sensitive block in the PLL. Two dominant topologies are used:

3-Stage Ring Oscillator and LC-Tank Resonator Diagrams



Ring Oscillator vs. LC-Tank VCO Comparison

Property	Ring Oscillator VCO	LC-Tank VCO
Structure	N cascaded inverter/delay stages in a ring	Parallel LC resonator with varactor for tuning
Phase Noise	Poor (-90 to -100 dBc/Hz at 1 MHz)	Excellent (-120 to -145 dBc/Hz at 1 MHz)
Tuning Range	Very wide ($> 2\times$)	Narrow (~ 10 – 20%)
Silicon Area	Very compact (inverters only)	Large (on-chip inductor is bulky)
Frequency	DC to > 10 GHz (digital CMOS)	500 MHz to > 60 GHz
Applications	Digital CDR, on-chip clocking	RF synthesizers, wireless transceivers

2. Delay-Locked Loop (DLL) Architecture & Working

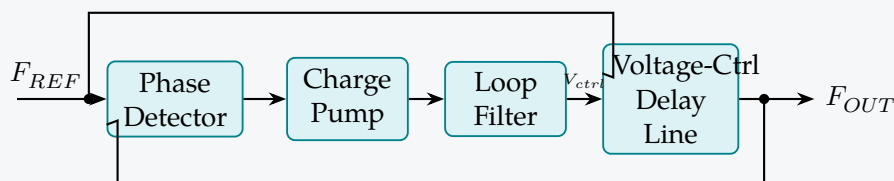
Delay-Locked Loop (DLL) Concept

A **Delay-Locked Loop (DLL)** is a feedback system that aligns the phase of an output clock with an input clock by controlling the delay of a Voltage-Controlled Delay Line (VCDL). Unlike a PLL, a DLL does not use an oscillator (VCO); it only shifts the phase of an existing clock.

2.1 Block Components and Signal Flow

A DLL consists of a Phase Detector (PD), a Charge Pump (CP), a Loop Filter (LF), and a Voltage-Controlled Delay Line (VCDL).

Block Diagram of a Delay-Locked Loop



2.2 PLL vs. DLL Stability & Jitter Comparison

Comparison of PLL and DLL Architectures

Parameter	Phase-Locked Loop (PLL)	Delay-Locked Loop (DLL)
Core Component	VCO (Voltage Controlled Oscillator)	VCDL (Voltage Controlled Delay Line)
System Order	Type-II, 2nd-order or higher (requires series R to stabilize)	Type-I, 1st-order system (unconditionally stable)
Jitter Behavior	Jitter Accumulation: Noise recirculates in the VCO tank/ring, accumulating over cycles	No Jitter Accumulation: Jitter is reset at the end of the delay line every cycle
Frequency Synthesis	Supports frequency multiplication ($F_{OUT} = NF_{REF}$)	Cannot multiply frequency directly; only shifts phase
Lock Range	Limited by VCO tuning range and loop dynamics	Limited by VCDL delay range; prone to harmonic locking

- **Stability Analysis:** The VCDL is a simple delay stage. Unlike a VCO, it does not integrate frequency to phase (it has no $1/s$ phase term). The DLL closed-loop system is first-order, meaning it is unconditionally stable and the loop filter requires only a single capacitor (no resistor R is needed).
- **Jitter Analysis:** In a VCO, if a clock edge is delayed due to noise, the subsequent edges remain shifted because the phase error accumulates over time. In a VCDL, the input clock is a clean external clock. Any noise-induced delay inside the VCDL affects only the current cycle; the next cycle starts with a fresh, clean reference edge.

3. Digital PLLs & Frequency Synthesizers

3.1 Digital PLL (DPLL) and All-Digital PLL (ADPLL)

Modern SOCs implement ADPLLs to avoid bulky analog capacitors and process scaling issues:

- **TDC:** A Time-to-Digital Converter replaces the PFD, converting phase differences directly into digital words.
- **Digital Loop Filter (DLF):** Replaces the analog loop filter, saving silicon area and allowing programmable tuning coefficients.
- **DCO:** A Digitally Controlled Oscillator replaces the VCO, using capacitor DAC banks to control oscillation frequency.

3.2 Frequency Synthesizers

Synthesizers generate multiple frequencies from a single reference:

- **Integer-N Synthesizer:** Output frequency is $F_{OUT} = NF_{REF}$. The step size is limited to F_{REF} .

- **Fractional-N Synthesizer:** The feedback divider toggles between N and $N + 1$ using a Delta-Sigma modulator. This allows the average division ratio to be fractional, providing fine frequency steps with a high reference frequency (reducing loop division noise).

Quick Revision — Module 5

PLLs & DLLs | PE-EC802A

Key Formulas at a Glance

VCO Output Frequency	$\omega_{out} = \omega_{fr} + K_{VCO}V_{ctrl}$
PFD/CP Gain	$K_d = \frac{I_{CP}}{2\pi}$
VCO Phase integration	$H_{VCO}(s) = \frac{K_{VCO}}{s}$
Loop Natural Frequency	$\omega_n = \sqrt{\frac{I_{CP}K_{VCO}}{2\pi NC_1}}$
Loop Damping Factor	$\zeta = \frac{R}{2} \sqrt{\frac{I_{CP}K_{VCO}C_1}{2\pi N}}$
Integer-N output	$F_{OUT} = NF_{REF}$

One-Line Definitions

- **VCO:** An oscillator whose output frequency is a linear function of its input control voltage.
- **PFD:** A phase-detector block that detects both phase and frequency differences, preventing false lock states.
- **Loop Filter:** A low-pass filter that filters out the high-frequency charge-pump ripple to generate a smooth control voltage.
- **Damping Factor:** A parameter that determines the stability and transient settling time of the PLL loop.
- **VCDL:** A line of buffer stages whose propagation delay is controlled by an input voltage.
- **ADPLL:** A phase-locked loop consisting entirely of digital components (TDC, Digital Filter, DCO).
- **Fractional-N Synthesizer:** A frequency synthesizer that achieves fractional division ratios by dynamically switching divider counts.

PLL vs. DLL Architecture Comparison

Property	Phase-Locked Loop (PLL)	Delay-Locked Loop (DLL)
Core block	VCO (integrates freq. to phase)	VCDL (shifts phase, no integration)
System order	2nd-order (Type-II), needs R to stabilize	1st-order, unconditionally stable
Jitter	Accumulates (VCO noise recirculates)	Non-accumulating (reset each cycle)
Freq. synthesis	Yes: $F_{out} = NF_{ref}$	No (phase shift only)
Lock range	Limited by VCO tuning range	Limited by VCDL delay range
Application	Frequency synthesis, clocking	Deskew, phase alignment

Frequently Asked / Viva Questions

Q1. What are the main blocks of a charge-pump PLL?

The five main blocks are the Phase Frequency Detector (PFD), Charge Pump (CP), Loop Filter (LF), Voltage-Controlled Oscillator (VCO), and the Feedback Divider ($/N$).

Q2. Why is a resistor needed in series with the capacitor in a PLL loop filter?

The VCO acts as an integrator ($1/s$). If the loop filter is a simple capacitor, it acts as a second integrator ($1/s$). This creates two poles at the origin, making the system unstable (phase margin is 0°). The resistor introduces a zero ($1 + sRC$) that restores stability.

Q3. Compare PLL and DLL stability.

A DLL is a first-order system because the VCDL only shifts phase and does not integrate frequency to phase. Therefore, it is unconditionally stable. A PLL is a second-order (or higher) system due to the VCO phase integration and loop filter, requiring careful stabilization.

Q4. Explain the difference in jitter behavior between a PLL and a DLL.

In a PLL, phase errors recirculate through the VCO feedback loop, leading to jitter accumulation over time. In a DLL, the input reference clock is clean, and the jitter is reset at the end of the VCDL line every clock cycle, preventing jitter accumulation.

Q5. What is a Phase Frequency Detector (PFD), and what is its advantage over a simple XOR Phase Detector?

An XOR PD only detects phase difference and can lock to harmonics or fail to lock if frequencies are different. A PFD detects both phase and frequency differences, ensuring the loop always locks to the correct frequency.

Q6. What is frequency multiplication in a PLL, and how is it achieved?

Frequency multiplication allows generating a high frequency output $F_{OUT} = NF_{REF}$ from a low frequency reference. It is achieved by placing a divide-by- N

Frequently Asked / Viva Questions (contd.)

counter in the feedback path, forcing the VCO to run N times faster to lock.

Q7. What is a Digitally Controlled Oscillator (DCO) and where is it used?

A DCO is an oscillator whose output frequency is controlled by digital inputs (typically selecting capacitors in a varactor DAC array). It is a core block in All-Digital PLLs (ADPLLs).

Q8. Explain the working of a fractional-N synthesizer.

A fractional-N synthesizer achieves a fractional division ratio (e.g. $N.25$) by dynamically toggling the feedback divider value between N and $N + 1$ (e.g., dividing by N for 75% of the time and $N + 1$ for 25% of the time). A Delta-Sigma modulator is used to shape the divider switching noise.

Q9. What is the function of the charge pump in a PLL?

It converts the digital UP and DN pulses from the PFD into analog current pulses, injecting or extracting charge from the loop filter to adjust the VCO control voltage.

Q10. What is a Time-to-Digital Converter (TDC)?

A TDC is a digital circuit that measures the time interval between two clock transitions and outputs it as a digital word. It replaces the analog PFD/CP in digital PLLs.